SEAMLESS: Specularity-Aware End-to-End Adaptive Masking and Luminance Estimation via Self-Supervision

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Abstract

Purpose: In Gastrointestinal (GI) endoscopy, 3D information about camera motion and the GI tract surface provides navigation cues to verify inspection completeness and revisit previous locations. However, accurate 3D understanding in this setting is challenging due to irregular illumination conditions created by a moving light source and specular reflections. This motivates developing pose and depth estimation models that are robust to these extreme illumination conditions.
Methods: We propose SEAMLESS, that similarly to previous approaches in this domain decomposes scene illumination from tissue albedo while estimating camera motion and scene depth. We improve previous state-of-the-art (SHADeS) with a fully self-supervised end-to-end illumination decomposition that decouples luminance, specularities, and scene albedo, removing prior reliance on hand-crafted specularity detectors. To the best of our knowledge, this is the first monocular depth model in this domain that explicitly learns to detect and separate specular reflections through light decomposition.
Results: SEAMLESS outperforms SHADeS and older baselines on every depth metric on phantom data (C3VD), reducing AbsRel from 0.1312 to 0.1104 relative to SHADeS. On the more realistic C3VDv2, which contains more specular distortion and residual debris, the margin widens (AbsRel 0.1423 to 0.1156), and on real EndoMapper colonoscopies SEAMLESS produces visibly cleaner depth.
Conclusion: SEAMLESS significantly improves depth estimation in the presence of specular distortions and environmental artifacts, and remains robust on realistic endoscopic data where existing methods fail.

Keywords: Monocular depth estimation, Self-supervised specular detection, Colonoscopy, Self-supervised learning, Intrinsic image decomposition

1 Introduction

In gastrointestinal (GI) endoscopy, estimation of the camera trajectory and reconstruction of the GI tract can support assisted navigation to verify inspection completeness [1], and help relocalize previously detected landmarks [2]. While recent advances in monocular 3D perception achieve increasingly more detailed 3D reconstructions of endoscopic scenes [3, 4], most of them work under assumptions of well lit scenes and simplified Lambertian light reflections. In endoscopy, these are often insufficient to model strong illumination effects caused by a moving directional light source and the abundance of specular reflections from wet tissue.

SHADeS [5] is a monocular depth estimation framework that partly alleviates illumination difficulties by decoupling endoscopic RGB images into three distinct elements: specular reflections, diffuse albedo, and luminance. Subsequently, depth and motion networks are trained via optimizing multi-view consistency of the albedo component, which encodes the tissue texture independent from illumination fluctuations and artifacts. Even though this architecture outperforms prior monocular depth methods on colonoscopy benchmarks, it depends on a single intensity-based specular reflection binary detector that is still vulnerable to errors. Simultaneously learning specular light detection with albedo/luminance decomposition, depth, and pose would enable overcoming the current limitations of SHADeS.

We present SEAMLESS (Specularity-Aware End-to-End Adaptive Masking and Luminance Estimation via Self-Supervision), a specular-aware monocular depth estimation framework targeting realistic endoscopic scenes. Our contributions are as follows:

- A self-supervised light decoupling model that can be trained end-to-end on real endoscopy video to decompose images into albedo, luminance, and specular reflection components. While previous work (SHADeS) models specularities as a pre-computed binary mask, we treat them as a learnable soft mask that can more accurately represent complex light effects.
- Incorporation of our light decomposition model within a self-supervised monocular pose and depth pipeline that validates geometry consistency subsequent to rectifying illumination corruptions with inpainting, thereby diminishing the quantity of artifacts in depth estimations.
- An evaluation of our method and prior work on multiple endoscopic datasets (C3VD, C3VDv2, EndoMapper). Notably, prior state-of-the-art is benchmarked for the first time on C3VDv2 which represents a more challenging dataset than its predecessor C3VD. Our method outperforms baselines across the different datasets, demonstrating notable differences in illumination-related challenges such as suppression of specular reflection artifacts and accurate shape recovery in dark regions around the lumen.

2 Related Work

Endoscopic depth estimation remains predominantly self-supervised, as ground-truth depth is rarely available. Contemporary efforts mitigate data scarcity through

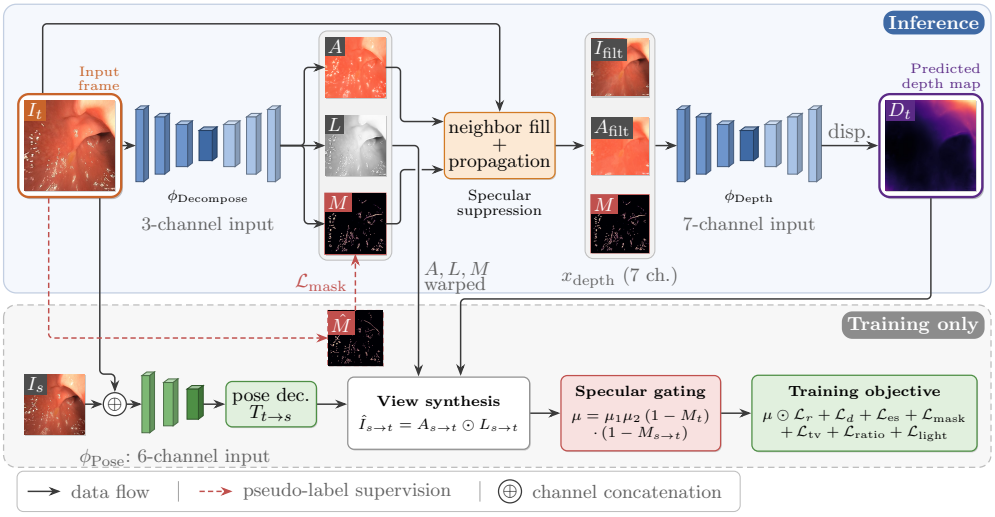


Fig. 1 Overview of SEAMLESS. $\phi_{\text{Decompose}}$ predicts the albedo A, shading L and soft specular mask M, supervised during training by the pseudo-label $\hat{M}$. M guides the suppression of highlights in the image and the albedo, which are concatenated with M as the input of ϕ_{Depth}. During training, the source decomposition is warped with the predicted depth and pose, and the photometric loss is gated by M in both views.

simulation-driven training and image translation [6], enhanced features for low-texture scenes [7], or foundation-model adaptation leveraging self-supervised intrinsics [8], yet all depend on photometric consistency, which breaks down under endoscopic illumination: specular reflections and non-Lambertian tissue impair both self-supervised and simulation-trained depth [9], while wide field-of-view endoscopes amplify these degradations [10].

EndoSLAM [11] demonstrated that monocular geometric inference is viable yet vulnerable to irregular photometric conditions, and AF-SfMLearner [12] introduced appearance flow for resilience against inconsistent illumination; however, neither model captures the interaction between albedo and illumination. IID-SfMLearner [13] (IID) derives the reprojection loss on intrinsically decomposed frames, though its Lambertian model lacks representative capacity for view-dependent highlights. SHADeS [5] contributes a non-Lambertian specular term (Sect. 3.1), but its specular handling remains external to the network (Sect. 1).

3 Methodology

3.1 Preliminaries

Self-supervised monocular depth. Given a target frame I_t and source frames I_s, $s \in \{t-1, t+1\}$, ϕ_{Depth} predicts the depth D_t of I_t and ϕ_{Pose} the camera motion $T_{t\to s}$. Using intrinsics K, every target pixel p_t maps into the source view, and sampling I_s at p_s yields reconstruction $\hat{I}_{s\to t}$, contrasted with I_t via the Monodepth2 photometric

loss [14] with $\alpha = 0.85$ [13]:

$$p_s \sim K\, T_{t\to s}\, D_t(p_t)\, K^{-1} p_t, \qquad \mathcal{L}_r(\hat{I}, I) = \alpha\, \frac{1 - \mathrm{SSIM}(\hat{I}, I)}{2} + (1-\alpha)\, \|\hat{I} - I\|_1. \quad (1)$$

View-dependent highlights on wet tissue violate the brightness constancy this loss assumes and introduce errors unrelated to geometry.

IID and SHADeS. IID [13] incorporates a decomposition network $\phi_{\text{Decompose}}$ factoring each frame into albedo A and shading L under the Lambertian model $I = AL$, plus a light adjustment network ϕ_{Adjust} predicting illumination offsets. SHADeS [5] employs the non-Lambertian model $I = AL + S$, yet S is not predicted by the network: a non-learned specular detector [15] and a pretrained inpainting model [16] generate $I_{\text{rem}} = I - S$, replacing I in the decomposition and reprojection losses, and a mask emerges only at inference by thresholding $I - AL$ at 50. Consequently, the network lacks internal representation of specularity, and detection errors remain fixed throughout training.

3.2 Overview of SEAMLESS

SEAMLESS moves specular processing into the network (Fig. 1). The decomposition network predicts an albedo A, a shading L and a soft specular mask M (Sect. 3.3), supervised via an annotation-free brightness-contrast pseudo-label $\hat{M}$ (Sect. 3.4). M serves to (i) inpaint the image alongside the albedo, which together with M (Sect. 3.5) constitute a seven-channel depth input; (ii) gate the photometric reprojection loss across the target and warped source views (Sect. 3.6); and (iii) increase depth smoothness inside highlights (Sect. 3.7). As a differentiable output optimized jointly with depth and decomposition, M operates without an external detector at inference.

3.3 Decomposition with a learned specular mask

Following SHADeS, the image is a diffuse plus a specular term, $I \approx A \odot L + S$, where $\odot$ is the element-wise product, A a three-channel albedo and L a single-channel shading. We do not estimate S; instead, $\phi_{\text{Decompose}}$ predicts a specular mask $M \in [0,1]^{H\times W}$, the per-pixel probability that a pixel is specular. All reconstructions use the diffuse term $A \odot L$ only, and M excludes the detected highlights from the losses and marks them for inpainting. $\phi_{\text{Decompose}}$ follows IID [13], with an ImageNet-pretrained ResNet-18 encoder and a U-Net decoder, plus a sigmoid-activated M head, whose bias is initialized to -2.0, beside the A and L heads, so the mask starts near $\sigma(-2) \approx 0.12$, a mostly non-specular prior. The single-scale network is shared across the three frames. Residual highlights in A are suppressed as $A_{\text{diff}} = A \odot (1 - M) + M \odot \bar{A}$, where $\bar{A}$ is the 7×7 average of A over non-specular neighbors; A_{diff} is used in the decomposition loss and in view synthesis, and is the starting point of the depth-input albedo.

3.4 Specular pseudo-labels

As no pixel-level specular annotation exists, M is supervised by a pseudo-label $\hat{M}$ computed from the raw target frame at every iteration and not used at inference.

Highlights appear as sharp local brightness peaks over smooth diffuse illumination, so from the brightness B (the RGB mean) we take the positive residual C over a 9×9 average, set it to zero outside a field-of-view region Ω, and select the pixels above its per-image 97th percentile $\tau_{0.97}$ as the core:

$$C = \max\big(B - \mathrm{AvgPool}_{9\times 9}(B),\, 0\big), \qquad \hat{M}_{\mathrm{core}}(p) = \mathbb{1}\big[C(p) > \tau_{0.97}\big]\, \mathbb{1}\big[p \in \Omega\big]. \quad (2)$$

Ω omits the high-contrast boundary of the endoscope vignette: it holds the pixels having $B > \tau_{\mathrm{black}} = 0.03$, eroded by 6 pixels, minus a 5% border margin on each side. Consequently the core becomes the top 3% of in-view pixels within every frame; across core ratios spanning from 0.5 to 15%, 3% produced the best overall outcome and the most credible masks upon visual inspection (Sect. 2 of the supplementary material). At the center of large saturated highlights the contrast approaches zero and the core isolates merely the rim, so unmasked pixels fully enclosed by the core, located via flood-filling from the image edge, are appended to $\hat{M}$; this fills enclosed specular regions without pushing the mask outside of the true specularity.

$\hat{M}$ anchors M to specularities instead of serving as a replication target. Within SHADeS, the detector output substitutes the image inside losses, causing its errors to persist throughout training; here $\hat{M}$ contributes solely via the weighted term $\mathcal{L}_{\mathrm{mask}}$, whereas M is likewise optimized end to end by the photometric loss, via the gate μ and the depth input, and by the decomposition loss across A_{diff}, allowing it to diverge from $\hat{M}$ whenever those objectives conflict (see the ablation in Sect. 3.7).

3.5 Specular-suppressed depth input

Highlights are unrelated to geometry yet spatially structured and locally dominant, so a depth encoder fed the raw image can learn spurious brightness-to-depth correlations. We therefore replace specular pixels with an estimate of the underlying diffuse appearance. For a map X, the average over a 7×7 window $\mathcal{N}(p)$ uses only non-specular pixels inside the field of view, so that the black vignette does not bias the fill:

$$\omega(q) = \big(1 - M(q)\big)\, \mathbb{1}\big[B(q) > \tau_{\mathrm{black}}\big], \qquad \bar{X}(p) = \frac{\sum_{q\in\mathcal{N}(p)} \omega(q)\, X(q)}{\sum_{q\in\mathcal{N}(p)} \omega(q) + \epsilon} \quad (3)$$

Since the image and the albedo diverge in brightness through the shading, the image fill rescales the neighbor-averaged albedo to match the standard BT.601 luma ℓ of the adjacent non-specular image, with $\gamma = \mathrm{clamp}\big(\ell(\bar{I})/(\ell(\bar{A}) + \epsilon), 0, 2\big)$ to avoid extreme color shifts, a fallback to $\bar{I}$ wherever the rescaled albedo proves invalid, and clipping at $[0, 1]$. Filtering the image alone is insufficient, because the albedo preserves a residual specular signal that resurfaces in depth, so the identical suppression applies to A_{diff} without rescaling, with $\bar{A}$ now serving as its valid-neighbor mean:

$$I_{\mathrm{filt}} = I \odot (1 - M) + \gamma\, \bar{A} \odot M, \qquad A_{\mathrm{filt}} = A_{\mathrm{diff}} \odot (1 - M) + \bar{A} \odot M. \quad (4)$$

Pixels within large highlights lacking valid neighbors are filled by 40 iterations of propagation, where a pixel adopts the weighted color of its known neighbors once their

weight exceeds 0.5; this value applies where $M > 0.5$, and Eq. 4 otherwise. The depth encoder obtains

$$x_{\text{depth}} = [\, I_{\text{filt}},\ A_{\text{filt}},\ M\,] \in \mathbb{R}^{7\times H\times W} \tag{5}$$

where I_{filt} denotes the specular-suppressed image, while A_{filt} represents the view-independent surface color, and M flags which pixels are unreliable.

3.6 Specular-aware view synthesis

ϕ_{Pose} follows Monodepth2 [14]: an ImageNet-pretrained ResNet-18 encoder takes a pair of frames, while a decoder regresses an axis-angle rotation and a translation, scaled by 0.1. For each source frame, its diffuse albedo, shading, and mask are warped into the target view via Eq. 1, and the reconstruction becomes their diffuse product, $\hat{I}_{s\to t} = A_{s\to t} \odot L_{s\to t}$, without the ϕ_{Adjust} of IID, following SHADeS. Comparison uses the raw target frame I_t, not an inpainted variant; instead of removing highlights from the target, they are excluded from the loss using the per-pixel weight

$$\mu = \mu_1 \odot \mu_2 \odot (1 - M_t) \odot (1 - M_{s\to t}), \tag{6}$$

where $\mu_2 = [\hat{I}_{s\to t} > 0]$ removes pixels warped from outside the source image [13], and $\mu_1 = [\min_s \mathcal{L}_r(\hat{I}_{s\to t}, I_t) < \min_s \mathcal{L}_r(I_s, I_t)]$ equals the auto-mask of Monodepth2 [14], evaluated on the raw frames. A pixel survives only when it appears non-specular in both views, causing a moving highlight to vanish from each. As M is soft, specular pixels contribute near-zero photometric gradient, leaving depth there controlled by the smoothness and mask components of Sect. 3.7.

3.7 Training objective

The full objective is

$$\mathcal{L} = \lambda_r\, \mu \odot \mathcal{L}_r + \lambda_d \mathcal{L}_d + \lambda_{\text{es}} \mathcal{L}_{\text{es}} + \lambda_{\text{m}} \mathcal{L}_{\text{mask}} + \lambda_{\text{tv}} \mathcal{L}_{\text{tv}} + \lambda_\rho \mathcal{L}_{\text{ratio}} + \lambda_{\text{light}} \mathcal{L}_{\text{light}} \tag{7}$$

Every term is evaluated solely at the finest scale. The decomposition loss, $\mathcal{L}_d = \frac{1}{|F|} \sum_{f\in F} \| I_f - A_{\text{diff},f} \odot L_f \|_1$, across the frames f of the triplet F, employs the raw image rather than an inpainted version, so decomposition no longer relies on an external pipeline. Only the ℓ_1 norm is used: SSIM matches local means, variances and cross-correlations, driving $A \odot L$ to replicate the local structure of I_f; this can be satisfied by allocating texture arbitrarily between A and L, while ℓ_1 enforces per-pixel agreement without that structural pressure. We also omit the albedo consistency loss of IID and SHADeS, which we found unnecessary. When the photometric signal is gated out, depth becomes underconstrained, so we increase the edge-aware smoothness of Monodepth2 [14] proportionally to the mask:

$$\mathcal{L}_{\text{es}} = \Big\langle \big(1 + \beta\, \text{sg}(M_t)\big) \Big(|\partial_x d_t|\, e^{-|\partial_x I_t|} + |\partial_y d_t|\, e^{-|\partial_y I_t|} \Big) \Big\rangle \tag{8}$$

where d_t is the predicted disparity (not mean-normalized), $\text{sg}(\cdot)$ stops the gradient, $\langle\cdot\rangle$ marks the spatial average, the edge term pulls from the color-augmented target frame,

and $\beta = 0.3$. The target mask aligns with the pseudo-label through binary cross-entropy, $\mathcal{L}_{\text{mask}} = \text{BCE}(M_t, \hat{M})$, the only link between $\hat{M}$ and the network; a total-variation term, $\mathcal{L}_{\text{tv}} = \langle |M_{i,j} - M_{i,j+1}| \rangle + \langle |M_{i,j} - M_{i+1,j}| \rangle$, penalizes speckled mask boundaries; and a coverage term, $\mathcal{L}_{\text{ratio}} = |\langle M_t \rangle - \rho|$ with $\rho = 0.15$, serves as an early-training stabilizer against mask collapse rather than as a sparsity objective. Finally, $\mathcal{L}_{\text{light}} = \langle |\partial_x L_t| \, e^{-|\partial_x I_t|} + |\partial_y L_t| \, e^{-|\partial_y I_t|} \rangle$ imposes the same edge-aware smoothness on L, preventing the shading from absorbing texture that belongs to the albedo. An ablation stripping each component individually (Sect. 3 of the supplementary material) validates all: dropping any component except $\mathcal{L}_{\text{mask}}$ raises AbsRel and lowers $\delta < 1.25$ on both C3VDv2 and EndoMapper. Removing $\mathcal{L}_{\text{mask}}$ marginally lifts depth on C3VD yet yields the poorest EndoMapper result on every metric, since M is detached from specularities and fails to transfer to real mucosa. We set $\lambda_r = 1.0$, $\lambda_d = 0.3$, $\lambda_{\text{es}} = 0.01$, $\lambda_{\text{m}} = 0.4$, $\lambda_{\text{tv}} = 0.005$, $\lambda_\rho = 0.02$ and $\lambda_{\text{light}} = 0.05$, where λ_r and λ_{es} follow IID [13].

4 Experiments

4.1 Implementation details

Training data. We train only on real colonoscopy video, employing the Hyper-Kvasir [17] split of SHADeS [5]: 16,976 training, 1,887 validation and 786 test frames drawn from videos containing Boston Bowel Preparation Scale 2 or 3, limited to 926 frames per clip. Following SHADeS, frames are center-cropped to a square, resized to 288×288 and undistorted with the C3VD [18] intrinsics, because Hyper-Kvasir provides none, and horizontal flipping is disabled; inputs are color-augmented, whereas photometric targets are not (details in the supplementary material). Training on real rather than phantom data yields better generalization [19], and C3VD, C3VDv2 and EndoMapper are never employed for training, so all reported results are zero-shot.

Optimization. We train for 20 epochs with Adam at an initial learning rate of 5×10^{-5}, decayed by a factor of 0.1 at epoch 10, with a batch size of 8 and triplets using frame offsets $\{0, -1, +1\}$, the first being the target. All encoders start from ImageNet weights except the first layer of the depth encoder, which is randomly initialized to accept the seven-channel input of Eq. 5; unlike SHADeS, we do not initialize from IID depth or Monodepth2 pose checkpoints. The final-epoch model is reported. All hyperparameters, optimizer configurations and hardware specifics appear in the supplementary material.

Baseline. For SHADeS, we use the original authors' checkpoint of the $\text{SHADeS}_{\text{IM,AM,NA}}$ configuration (inpainting module, auto-masking, no adjustment network), trained on identical Hyper-Kvasir data at 288×288 [5].

4.2 Evaluation protocol

Depth on C3VD. As in SHADeS, we evaluate on all frames of C3VD [18]. Each prediction is median-scaled to the ground truth and then clipped to $[10^{-3}, 150]$; only ground-truth pixels inside that interval are assessed. We report MAE, MedAE, RMSE, $\text{RMSE}_{\log}$, AbsRel, SqRel and $\delta < 1.25^k$ for $k = 1, 2, 3$, with MAE, MedAE, RMSE and SqRel expressed in millimeters. For comparison with prior work on this benchmark,

we also report the results of Monodepth2 [14], MonoViT [20], IID [13] and SHADeS, as published in [5].

Depth on C3VDv2. Evaluation covers all frames of C3VDv2 [21], applying equivalent scaling, clipping, and metrics as for C3VD. No published results exist on C3VDv2 for the comparative methods, so the original SHADeS checkpoint (Sect. 4.1) is assessed under the identical protocol.

Depth on EndoMapper. EndoMapper [22] provides COLMAP reconstructions of real colonoscopies, which we use as sparse pseudo ground truth. Frames are undistorted with the provided calibration before inference, and each prediction is resized to the native resolution, sampled at the visible COLMAP keypoints and median-scaled per frame; metrics are averaged over frames. After discarding invalid points and frames with too few of them (details in the supplementary material), we evaluate 14,213 frames from 119 sequences.

5 Results and Discussion

C3VD. Table 1 compares SEAMLESS against the self-supervised methods benchmarked on C3VD in [5]. SEAMLESS attains the top result on every metric. Relative to the leading prior method, SHADeS, it reduces AbsRel by 16% (0.1312 to 0.1104) and MAE from 4.4 mm to 3.7 mm, an AbsRel improvement roughly three times that of SHADeS over Monodepth2 (0.0208 versus 0.0072).

The visualization in Fig. 2 presents depth predicted on C3VD frames. Compared to SHADeS, SEAMLESS renders the overall geometry of the colon with greater fidelity, aligning with the consistent improvements seen in Table 1.

C3VDv2. On C3VDv2, whose sequences add fecal debris, mucous pools, blood and fast camera motion [21], SEAMLESS again outperforms SHADeS on every metric (Table 1), by a larger margin than on C3VD: AbsRel drops by 19% (0.1423 to 0.1156) and $\delta < 1.25$ climbs by 5.4 points. From C3VD to C3VDv2, the AbsRel of SHADeS grows roughly twice as much as that of SEAMLESS (0.0111 versus 0.0052), so SEAMLESS is less affected by the added realism. Qualitatively, SEAMLESS recovers internal details in the deep regions of the lumen with sharper definition and higher confidence (Fig. 3a).

EndoMapper. On EndoMapper, SEAMLESS outperforms SHADeS on every metric, but by small margins (AbsRel 1.1% lower), which we do not consider decisive. The COLMAP reference originates from sparse triangulation on low-texture, wet mucosa and contains unknown error, consistent with an AbsRel near 0.38 for both models, and its keypoints focus on textured, non-specular regions where specular processing is least critical. EndoMapper errors are expressed in arbitrary COLMAP units and cannot be compared with C3VDv2. The qualitative contrast is far more pronounced (Fig. 3b): SHADeS fails severely, mispredicting the geometry completely and generating false dark patches in the depth map, whereas SEAMLESS remains stable and plausible. SEAMLESS suppresses specular regions prior to depth encoding (Sect. 3.5) and omits them from the photometric loss (Sect. 3.6), while the depth network of SHADeS receives them untouched. Since these breakdowns appear only in a subset of frames

Table 1 Depth estimation results on C3VD and C3VDv2 (in mm) and on 119 EndoMapper sequences against sparse COLMAP pseudo ground truth (in COLMAP units). On C3VD, results of all methods except SEAMLESS are reported from [5]. δ_k denotes $\delta < 1.25^k$. Lower is better for error metrics and higher for δ_k. Best result per dataset in bold.

Method	MAE	MedAE	RMSE	$\text{RMSE}_{\log}$	AbsRel	SqRel	δ_1	δ_2	δ_3
C3VD									
Monodepth2	4.6	3.3	6.3	0.1694	0.1384	1.0198	0.8114	0.9839	0.9987
MonoViT	5.0	3.1	7.4	0.1667	0.1391	1.2823	0.8230	0.9839	0.9991
IID	4.6	3.2	6.8	0.1856	0.1476	1.2793	0.8053	0.9613	0.9926
$\text{SHADeS}_{\text{IM,AM,NA}}$	4.4	3.1	6.3	0.1609	0.1312	0.9599	0.8397	0.9858	0.9985
SEAMLESS	**3.7**	**2.5**	**5.2**	**0.1417**	**0.1104**	**0.6615**	**0.8753**	**0.9900**	**0.9993**
C3VDv2									
SHADeS	5.2	3.9	7.1	0.1756	0.1423	1.2120	0.7897	0.9753	0.9985
SEAMLESS	**4.3**	**3.2**	**5.7**	**0.1471**	**0.1156**	**0.7695**	**0.8439**	**0.9887**	**0.9996**
EndoMapper									
SHADeS	7.9	4.9	12.0	0.4118	0.3803	15.2570	0.6044	0.8268	0.9099
SEAMLESS	**7.7**	**4.8**	**11.7**	**0.4090**	**0.3763**	**14.7141**	**0.6058**	**0.8279**	**0.9123**

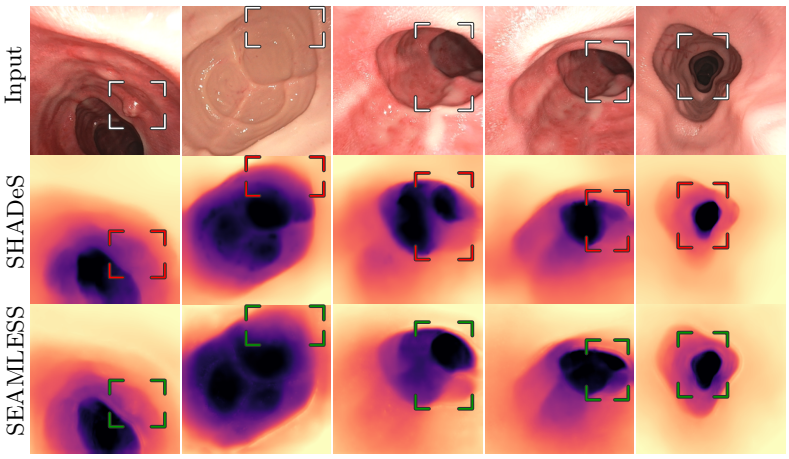


Fig. 2 Qualitative depth comparison on C3VD. Rows show the input frame and the depth predicted by SHADeS and SEAMLESS (darker is farther).

and are only sparsely captured by the COLMAP keypoints, the frame-averaged metrics understate their impact.

Pose. SHADeS leads in relative pose on EndoMapper (rotation error 0.408° vs. 0.447°), but SEAMLESS leads on every pose metric on C3VD and C3VDv2 (supp. Sects. 5–6); depth is predicted without ϕ_{Pose}.

Limitations. The real-data evaluation relies on sparse COLMAP pseudo ground truth of unknown accuracy, biased toward textured, non-specular regions, so our EndoMapper conclusions rest mainly on qualitative evidence. Besides the mask, SEAMLESS uses an ℓ_1-only decomposition loss against the raw image, drops the albedo consistency

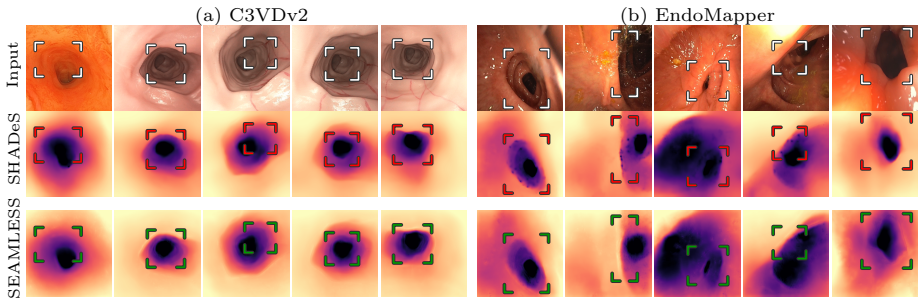


Fig. 3 Qualitative depth comparison on (a) C3VDv2 and (b) EndoMapper. Rows show the input frame and the depth predicted by SHADeS and SEAMLESS (darker is farther).

term, and differs from SHADeS in initialization and pose output scale; these changes have not been ablated in isolation, so the gains cannot yet be attributed to the mask alone, and they may explain the weaker pose estimates on EndoMapper. The mask also over-detects along fold boundaries where depth changes sharply (Sect. 2.1 of the supplementary material), so these pixels lose photometric supervision and rely on the smoothness prior. Finally, training uses only Hyper-Kvasir and temporal consistency is not evaluated.

6 Conclusion

We introduced SEAMLESS, which replaces the external specular detection and inpainting of SHADeS with a soft specular mask learned within the decomposition network from annotation-free pseudo-labels, employing it to dampen highlights in the depth input, gate the photometric loss and strengthen depth smoothness, thereby training the pipeline end to end. When trained solely on real video, SEAMLESS outperforms all competing methods on every depth metric on C3VD, expands its margin over SHADeS on the more realistic C3VDv2, and yields visibly cleaner depth on real EndoMapper colonoscopies. As reliable depth under specular distortion is a prerequisite for the coverage estimation and relocalization that motivate 3D reconstruction in colonoscopy, this robustness on real data is a step toward practical navigation support. Future work will ablate the remaining design changes in isolation, evaluate against denser real-world reference geometry, and broaden the mask to additional non-geometric artifacts like debris and fluid.

Declarations

Funding. No funding was received. **Competing interests.** The authors declare no competing interests. **Ethics approval and consent.** Not applicable; only publicly available, de-identified datasets [17, 18, 21, 22] were used.
Data, code and/or material availability. All datasets are publicly available from their original sources (EndoMapper on request through Synapse). Code is available from the corresponding author on reasonable request.

References

[1] Ma, R., Wang, R., Pizer, S., Rosenman, J., McGill, S.K., Frahm, J.-M.: Real-time 3d reconstruction of colonoscopic surfaces for determining missing regions. In: Medical Image Computing and Computer Assisted Intervention – MICCAI 2019. Lecture Notes in Computer Science, vol. 11768. Springer, Cham (2019)

[2] Morlana, J., Tardós, J.D., Montiel, J.M.M.: ColonMapper: topological mapping and localization for colonoscopy. In: IEEE International Conference on Robotics and Automation (ICRA) (2024)

[3] Bonilla, S., Zhang, S., Psychogyios, D., Stoyanov, D., Vasconcelos, F., Bano, S.: Gaussian pancakes: Geometrically-regularized 3D Gaussian splatting for realistic endoscopic reconstruction. In: Medical Image Computing and Computer Assisted Intervention – MICCAI 2024, pp. 274–283. Springer, Cham (2024)

[4] Gómez Rodríguez, J.J., Montiel, J.M.M., Tardós, J.D.: NR-SLAM: Non-rigid monocular SLAM. IEEE Transactions on Robotics **40**, 4252–4264 (2024)

[5] Daher, R., Vasconcelos, F., Stoyanov, D.: Shades: self-supervised monocular depth estimation through non-lambertian image decomposition. International Journal of Computer Assisted Radiology and Surgery **20**(6), 1255–1263 (2025)

[6] Jeong, B.H., Kim, H.K., Son, Y.D.: Depth estimation from monocular endoscopy using simulation and image transfer approach. Computers in biology and medicine **181**, 109038 (2024)

[7] He, Q., Feng, G., Bano, S., Stoyanov, D., Zuo, S.: Monolot: Self-supervised monocular depth estimation in low-texture scenes for automatic robotic endoscopy. IEEE Journal of Biomedical and Health Informatics **28**(10), 6078–6091 (2024)

[8] Cui, B., Islam, M., Bai, L., Wang, A., Ren, H.: Endodac: Efficient adapting foundation model for self-supervised depth estimation from any endoscopic camera. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 208–218 (2024). Springer

[9] Liu, X., Sinha, A., Ishii, M., Hager, G.D., Reiter, A., Taylor, R.H., Unberath, M.: Dense depth estimation in monocular endoscopy with self-supervised learning methods. IEEE transactions on medical imaging **39**(5), 1438–1447 (2019)

[10] Mathew, A., Magerand, L., Trucco, E., Manfredi, L.: Self-supervised monocular depth estimation for high field of view colonoscopy cameras. Frontiers in Robotics and AI **10**, 1212525 (2023)

[11] Ozyoruk, K.B., Gokceler, G.I., Bobrow, T.L., Coskun, G., Incetan, K., Almalioglu, Y., Mahmood, F., Curto, E., Perdigoto, L., Oliveira, M., *et al.*: Endoslam dataset and an unsupervised monocular visual odometry and depth estimation approach

for endoscopic videos. Medical image analysis **71**, 102058 (2021)

[12] Shao, S., Pei, Z., Chen, W., Zhu, W., Wu, X., Sun, D., Zhang, B.: Self-supervised monocular depth and ego-motion estimation in endoscopy: Appearance flow to the rescue. Medical image analysis **77**, 102338 (2022)

[13] Li, B., Liu, B., Zhu, M., Luo, X., Zhou, F.: Image intrinsic-based unsupervised monocular depth estimation in endoscopy. IEEE Journal of Biomedical and Health Informatics **29**(12), 9084–9094 (2024)

[14] Godard, C., Mac Aodha, O., Firman, M., Brostow, G.J.: Digging into self-supervised monocular depth estimation. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 3828–3838 (2019)

[15] El Meslouhi, O., Kardouchi, M., Allali, H., Gadi, T., Benkaddour, Y.A.: Automatic detection and inpainting of specular reflections for colposcopic images. Central European Journal of Computer Science **1**(3), 341–354 (2011)

[16] Daher, R., Vasconcelos, F., Stoyanov, D.: A temporal learning approach to inpainting endoscopic specularities and its effect on image correspondence. Medical Image Analysis **90**, 102994 (2023)

[17] Borgli, H., Thambawita, V., Smedsrud, P.H., Hicks, S., Jha, D., Eskeland, S.L., Randel, K.R., Pogorelov, K., Lux, M., Nguyen, D.T.D., *et al.*: Hyperkvasir, a comprehensive multi-class image and video dataset for gastrointestinal endoscopy. Scientific data **7**(1), 283 (2020)

[18] Bobrow, T.L., Golhar, M., Vijayan, R., Akshintala, V.S., Garcia, J.R., Durr, N.J.: Colonoscopy 3d video dataset with paired depth from 2d-3d registration. Medical Image Analysis **90**, 102956 (2023)

[19] Ju, X., Daher, R., Stoyanov, D., Bano, S., Vasconcelos, F.: Multi-modal monocular endoscopic depth and pose estimation with edge-guided self-supervision. International Journal of Computer Assisted Radiology and Surgery, 1–9 (2026)

[20] Zhao, C., Zhang, Y., Poggi, M., Tosi, F., Guo, X., Zhu, Z., Huang, G., Tang, Y., Mattoccia, S.: MonoViT: Self-supervised monocular depth estimation with a vision transformer. In: International Conference on 3D Vision (3DV), pp. 668–678 (2022)

[21] Golhar, M.V., Galeano Fretes, L.S., Ayers, L., Akshintala, V.S., Bobrow, T.L., Durr, N.J.: C3VDv2: Colonoscopy 3D video dataset with enhanced realism. arXiv preprint arXiv:2506.24074 (2025)

[22] Azagra, P., Sostres, C., Ferrández, Á., Riazuelo, L., Tomasini, C., Barbed, O.L., *et al.*: Endomapper dataset of complete calibrated endoscopy procedures. Scientific Data **10**, 671 (2023)